

CTBC: Contact-Triggered Blind Climbing for Wheeled Bipedal Robots with Instruction Learning and Reinforcement Learning

Rankun Li^{*,1,2}, Hao Wang^{*,2}, Qi Li^{*,2}, Zhuo Han^{1,2}, Yifei Chu², Linqi Ye^{†,1}, Wende Xie^{†,2}, and Wenlong Liao²

Abstract—In recent years, wheeled bipedal robots have garnered significant attention due to their exceptional mobility on flat terrain. However, while stair climbing has been achieved in prior studies, these existing methods often suffer from a severe lack of versatility, making them difficult to adapt to varying hardware specifications or diverse complex terrains. To overcome these limitations, we propose a generalized Contact-Triggered Blind Climbing (CTBC) framework. Upon detecting wheel-obstacle contact, the framework triggers a leg-lifting motion integrated with a strongly-guided feedforward trajectory. This allows the robot to rapidly acquire agile climbing skills, significantly enhancing its capability to traverse unstructured environments. Distinct from previous approaches, CTBC demonstrates superior robustness and adaptability, having been validated across multiple wheeled bipedal platforms with different wheel radii and tire materials. Real-world experiments demonstrate that, relying solely on proprioceptive feedback, the proposed framework enables robots to achieve reliable and continuous climbing over obstacles well beyond their wheel radius. **Project page:** <https://ctbc-for-wheeled-bipedal-robots.github.io/>

Index Terms—Contact-Based Control, Instruction Learning, Reinforcement Learning, Wheeled Bipedal Robots.

I. INTRODUCTION

TRADITIONAL legged robots have demonstrated remarkable agility and adaptability on complex terrain, yet their locomotion efficiency and speed remain comparatively low [1], making it difficult to satisfy the demands of rapid mobility. Wheeled-legged robots, benefiting from their high energy efficiency over long distances and superior travel speed [2], have been extensively studied and deployed across various domains. Nevertheless, when confronted with challenging environments such as staircases or uneven surfaces, the inherent limitations of wheeled robots become evident, as they lack the flexibility required to surmount obstacles effectively.

For wheeled-legged robots, tire dimensions exert a decisive influence on the feasibility of stair-climbing. Larger wheels confer a clear advantage; for instance, Simon Chamorro et al. [3] demonstrated that the Ascento robot can ascend a 15 cm stair with wheels of 25 cm radius, but only with deflated tires and limited to climbing a single step.

To overcome the pain points of wheeled-legged robots in complex terrain, we draw inspiration from the contact-triggered reflexes observed in human gait and propose a contact-triggered control framework that synergizes feedforward instruction learning with reinforcement learning. By



Fig. 1. We have developed a contact-triggered blind climbing control policy that works for wheeled-legged robots of various tire sizes, enabling them to conquer a variety of challenging terrains.

leveraging privileged information within an asymmetric actor-critic architecture, we develop a biomimetic gait reflex that enables wheeled-biped robots to ascend stairs with ease. In contrast, our method is suitable for wheeled-biped robots with arbitrary wheel diameters and tire types, and has been successfully deployed on robots with 11 cm rubber solid tires and 12.7 cm pneumatic tires. The key contributions of this work are summarized as follows:

- 1) **Contact-Triggered RL Task Framework.** We propose a contact-triggered reinforcement learning task framework applicable to wheeled-legged robots of various tire sizes, including the small-tire robot *LimX Dynamics Tron1*, to achieve stair-climbing strategies.
- 2) **Feedforward Trajectory Instruction Learning.** By combining instruction learning with reinforcement learning, feedforward trajectories are utilized to teach the robot when to lift its legs appropriately. This approach avoids unnecessary exploration and significantly improves learning efficiency.
- 3) **A Universal Framework for Various Wheel Sizes.** This method enables the *LimX Dynamics Tron1* robot to continuously climb 20 cm stairs (Fig. 1), far exceeding its tire radius. It also enables the *Cowarobot R0* heavy-duty wheeled-biped robot to continuously climb 7.5 cm stairs. The strategy supports both fast movement on flat ground and motion on complex terrain, enhancing the practical adaptability of the robot.

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CTBC: 面向轮式双足机器人的接触触发式盲攀爬方法，结合指令学习与强化学习

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摘要—近年来，轮式双足机器人凭借其在平坦地形上的卓越机动性而受到广泛关注。然而，尽管已有研究实现了爬楼梯功能，这些现有方法往往缺乏足够的通用性，难以适应不同的硬件配置或多样的复杂地形。为克服这些局限，我们提出了一种通用的接触触发式盲攀爬（CTBC）框架。在检测到轮-障碍物接触后，该框架触发抬腿动作，并结合强引导的前馈轨迹。这使得机器人能够快速习得敏捷的攀爬技能，显著增强其在非结构化环境中的通行能力。与以往方法不同，CTBC展现出优越的鲁棒性和适应性，已在多种具有不同车轮半径和轮胎材料的轮式双足平台上得到验证。真实世界实验表明，仅依靠本体感觉反馈，所提出的框架即可使机器人实现对其车轮半径远超障碍物的可靠、连续攀爬。项目页面：

<https://ctbc-for-wheeled-bipedal-robots.github.io/>

索引词——基于接触的控制、指令学习、强化学习、轮式双足机器人。

TRADITIONAL 腿式机器人在复杂地形上展现了卓越的灵活性和适应性，但其运动效率和速度仍然相对较低，[1]难以满足快速移动的需求。轮腿机器人凭借其在长距离上的高能量效率和优越的行驶速度，[2]，已在各个领域得到广泛研究和应用。然而，当面对楼梯或不平整表面等具有挑战性的环境时，轮式机器人的固有局限性便显现出来，因为它们缺乏有效克服障碍所需的灵活性。

对于轮腿机器人而言，轮胎尺寸对爬楼梯的可行性具有决定性影响。较大的轮子具有明显优势；例如，Simon Chamorro 等人 [3] 证明，Ascento 机器人可以使用半径为 25 厘米的轮子爬上 15 厘米高的台阶，但前提是轮胎必须放气，且仅限于攀爬单个台阶。

为克服轮腿机器人在复杂地形中的痛点，我们从人类步态中观察到的接触触发反射中汲取灵感，提出了一种将前馈指令学习与强化学习协同结合的接触触发式控制框架。通过

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图1. 我们开发了一种适用于不同轮胎尺寸轮腿机器人的接触触发式盲攀爬控制策略，使其能够征服各种具有挑战性的地形。

利用非对称演员-评论家架构中的特权信息，我们开发了一种仿生步态反射，使轮式双足机器人能够轻松攀爬楼梯。相比之下，我们的方法适用于任意轮径和轮胎类型的轮式双足机器人，并已成功部署在配备 11 厘米橡胶实心轮胎和 12.7 厘米充气轮胎的机器人上。本工作的主要贡献总结如下：

- 1) **接触触发强化学习任务框架。**我们提出了一种适用于不同轮胎尺寸轮腿机器人的接触触发强化学习任务框架，包括小轮胎机器人 *LimX Dynamics Tron1*，以实现爬楼梯策略。
- 2) **前馈轨迹指令学习。**通过将指令学习与强化学习相结合，利用前馈轨迹来教导机器人何时适当地抬腿。这种方法避免了不必要的探索，显著提高了学习效率。
- 3) **适用于不同轮径的通用框架。**该方法使 *LimX Dynamics Tron1* 机器人能够持续攀爬 20 厘米高的楼梯（图1），远超其轮胎半径。同时，该方法也使 *Cowarobot R0* 重型轮式双足机器人能够持续攀爬 7.5 厘米高的楼梯。该策略既支持在平坦地面上的快速移动，也支持在复杂地形上的运动，增强了机器人的实际适应性。

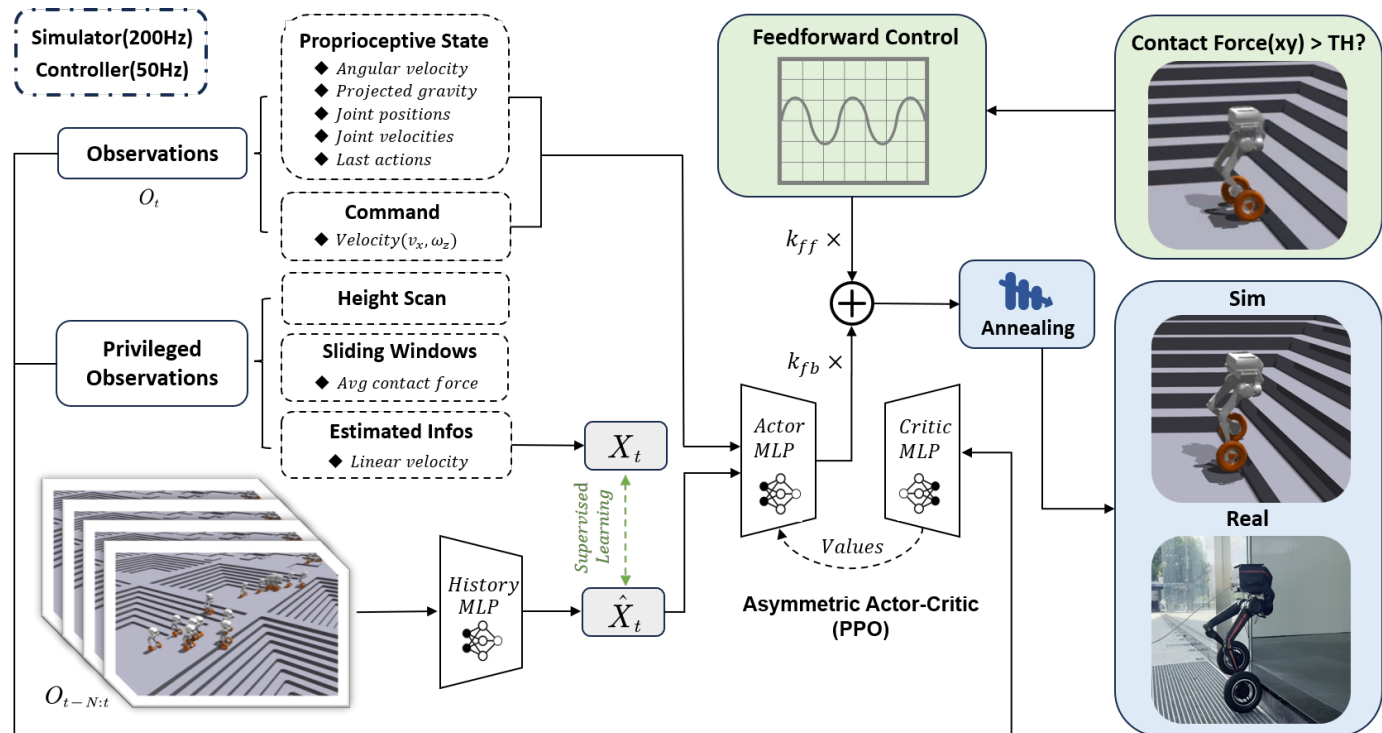


Fig. 2. Overview of our universal contact-triggered blind stair-climbing framework. The overall framework is mainly composed of a state estimator and an asymmetric actor-critic network. For elastic tires, we introduce a contact force sliding window to simulate more realistic contact. When the contact force in the xy direction of the foot exceeds a threshold, it triggers the designed feedforward reference trajectory to guide the robot to lift its leg. After annealing the feedforward trajectory, the method can be zero-shot transferred to the physical robot.

II. RELATED WORK

A. RL-Based Legged Locomotion

In the realm of legged locomotion, model-based optimization techniques such as Model Predictive Control (MPC) and trajectory optimization have long been widely adopted [4]. Nevertheless, the reliance of these methods on accurate and intricate dynamic models often constrains their robustness and generalization. Recently, model-free reinforcement learning has risen to prominence, offering an end-to-end learning paradigm that simultaneously strengthens robustness and markedly enhances generalization across diverse motion-control tasks, positioning itself as a powerful alternative to traditional approaches.

In 2019, Hwangbo et al. [5] introduced reinforcement learning to legged robot control by proposing a policy conditioned on desired velocity that outputs joint-position targets, augmented with an actuator network to enhance motor modeling accuracy. Building directly on this seminal framework, Lee et al. [6] developed a teacher-student architecture that enables robust traversal of hills, steps, and other challenging terrains using only on-board proprioception. Siekmann et al. [7] further demonstrated that an RL policy trained solely on proprioception can drive the bipedal robot Cassie to blindly ascend real-world stairs. Leveraging massively GPU-parallel simulation in Isaac Gym, Rudin et al. [8] trained a locomotion policy with Proximal Policy Optimization (PPO) in just 20 minutes that transfers zero-shot to real hardware. Extending this promising work, Rudin et al. [9] introduced a task formulation based

on positional goals: within a strict time limit, the robot must reach a target location, autonomously planning both path and motion to overcome obstacles and complete navigation without additional motion priors.

B. Wheeled-Legged Locomotion on Rough Terrain

In complex-terrain locomotion, conventional model-based methods often hinge on either simple heuristics that dictate when to walk or when to drive [10], or on fixed, pre-defined gait sequences [11]. Most policies for legged robots still embed hand-engineered gait patterns [12], [13] or biologically inspired motion primitives [6], [14].

Bjelonic et al. [15] introduced an online gait generator driven by leg availability: when the availability score of any leg drops below a threshold, the leg is automatically switched to swing while the others continue to drive or support; a single MPC parameter set suffices for all gaits, eliminating manual cost-weight tuning. Klemm et al. [16] leveraged non-smooth trajectory optimization to co-solve global motion planning and contact switching for stairs, steps and jumps in one pass, creating a closed perception-control loop and demonstrating continuous stair climbing on the Ascento wheeled-leg platform. Lee et al. [17] trained a quadrupedal wheeled robot with RL to switch on-the-fly between high-speed wheel driving and legged obstacle clearance in response to commands and terrain, enabling robust obstacle traversal. Lee et al. [18] further proposed a fully-integrated end-to-end framework that fuses model-free RL, privileged learning and hierarchical control, allowing seamless transitions between walking and driving for

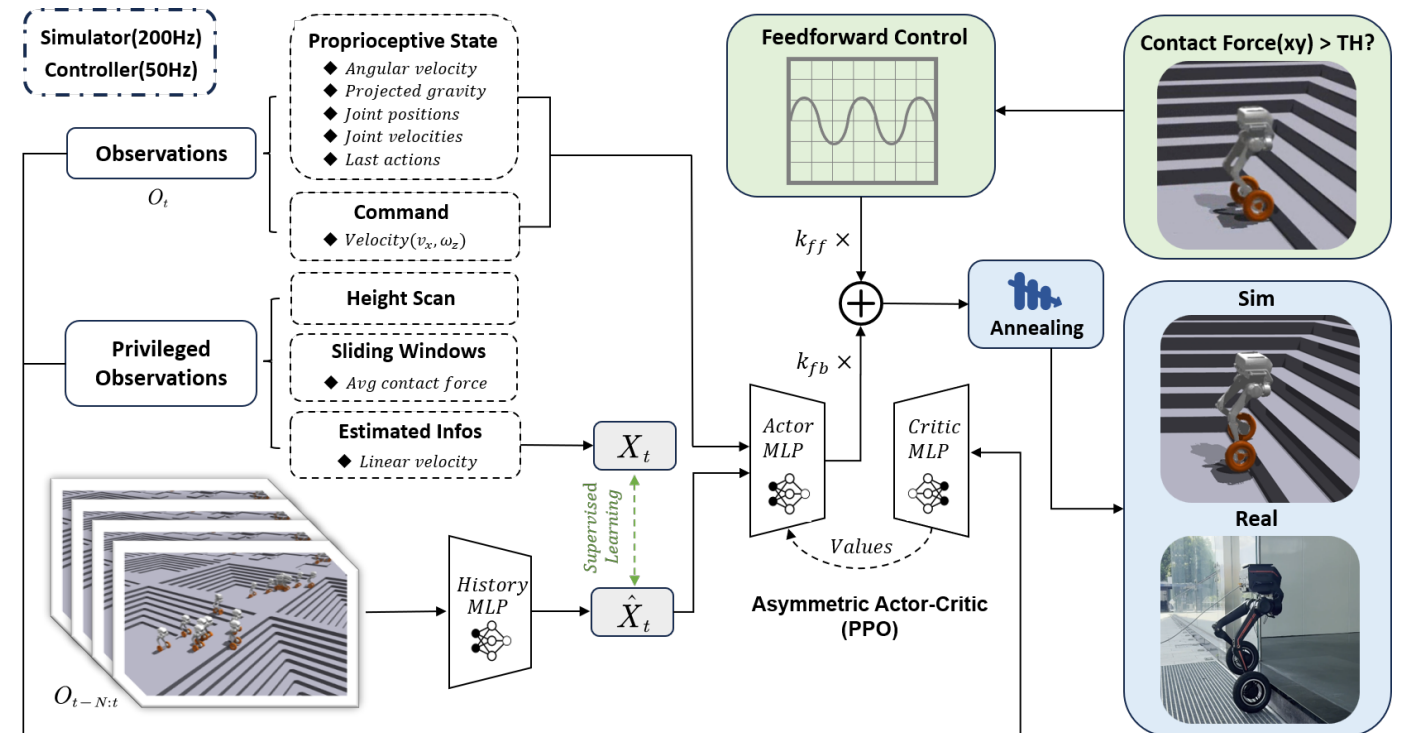


图2. 我们通用的接触触发式盲爬楼梯框架概览。该整体框架主要由状态估计器和非对称演员-评论家网络组成。针对弹性轮胎，我们引入了接触力滑动窗口来模拟更真实的接触。当足部在xy方向上的接触力超过阈值时，会触发设计好的前馈参考轨迹，引导机器人抬腿。在前馈轨迹退火后，该方法可以零样本迁移到物理机器人上。

II. 相关工作

A. 基于强化学习的腿式运动

在腿部运动领域，基于模型的优化技术，如模型预测控制（MPC）和轨迹优化，长期以来一直被广泛采用 [4]。然而，这些方法对精确且复杂的动力学模型的依赖往往限制了其鲁棒性和泛化能力。近年来，无模型强化学习逐渐崭露头角，提供了一种端到端学习范式，在增强鲁棒性的同时显著提升了跨多种运动控制任务的泛化能力，成为传统方法的有力替代方案。

2019年，Hwangbo等人 [5]通过提出一种以期望速度为条件的策略，将强化学习引入腿式机器人控制领域，该策略输出关节位置目标，并辅以执行器网络以提升电机建模精度。在这一开创性框架的基础上，Lee等人 [6]开发了教师-学生架构，使得机器人仅依靠机载本体感觉即可稳健地穿越丘陵、台阶及其他具有挑战性的地形。Siekmann等人 [7] 进一步证明，仅基于本体感觉训练的强化学习策略能够驱动双足机器人Cassie盲目攀爬真实世界的楼梯。借助Isaac Gym中的大规模GPU并行仿真，Rudin等人 [8]使用近端策略优化（PPO）在短短20分钟内训练出一种运动策略，该策略可零样本迁移至真实硬件。在这一前景广阔的工作基础上，Rudin等人 [9] 提出了一种基于位置目标的任务表述，

即在严格的时间限制内，机器人必须到达目标位置，自主规划路径和运动以克服障碍并完成导航，而无需额外的运动先验。

B. 粗糙地形上的轮腿运动

在复杂地形运动中，传统的基于模型的方法通常依赖于简单的启发式规则来决定何时行走或何时驱动 [10]，或依赖于固定的、预定义的步态序列 [11]。大多数腿式机器人的策略仍然嵌入了手工设计的步态模式 [12], [13]或生物启发的运动基元 [6], [14]。

Bjelonic 等人 [15] 提出了一种由腿部可用性驱动的在线步态生成器：当任何腿部的可用性评分低于阈值时，该腿自动切换至摆动相，而其他腿继续驱动或支撑；仅需一组 MPC参数集即可适用于所有步态，无需手动调整成本权重。Klemm 等人 [16] 利用非光滑轨迹优化，在一次求解中同时解决全局运动规划和接触切换问题，适用于楼梯、台阶和跳跃，构建了闭环感知-控制回路，并在Ascento轮腿平台上展示了连续爬楼梯能力。Lee 等人 [17]使用强化学习训练四足轮式机器人，使其能够根据指令和地形在高速轮式驱动与腿部障碍物清除之间实时切换，从而实现稳健的障碍物穿越。Lee 等人 [18]进一步提出了一种全集成端到端框架，融合了无模型强化学习、特权学习和分层控制，实现了行走与驱动之间的无缝切换，适用于

tasks such as table jumping and stair climbing. Chamorro et al. [3] demonstrated that a blind RL policy, operating without vision or localization, allows the Ascento robot to climb 15 cm stairs by relying solely on positional objectives, binary terrain flags, and an asymmetric actor-critic architecture. However, to the best of our knowledge, while the Ascento robot possesses a wheel diameter of 25 cm, both its simulation and hardware experiments only demonstrate the traversal of a single-step stair and lack the capability to ascend stairs with narrow treads.

Currently, a universal obstacle-traversal framework for bipedal wheeled robots with arbitrary wheel sizes remains elusive. In particular, when a robot is required to surmount high steps exceeding its wheel radius without any additional exteroceptive sensing, existing methods often struggle with performance and training convergence.

III. METHODOLOGY

As illustrated in Fig. 2, our universal contact-triggered blind climbing framework is depicted. The following sections systematically detail the training environment, reinforcement-learning task formulation, training pipeline, design of the contact-triggered mechanism, integration of feedforward instruction learning, and the sim-to-real transfer strategy with concrete deployment specifics.

A. Learning Environment

1) *Simulator*: We select Isaac Gym [19] as our training platform because it is specifically designed for reinforcement-learning applications and is equipped with a GPU-accelerated architecture that dramatically increases agent-training speed thanks to its high degree of parallelism. In addition, Isaac Gym supports domain randomization techniques [20], which improve the robustness of reinforcement-learning agents by introducing environmental variations during training and thus facilitate the transfer of policies to the real world. To validate our sim-to-real pipeline, we also leverage MuJoCo [21] for cross-validation of the trained policies [22]. MuJoCo is renowned for its high-fidelity simulation and is widely used to verify reinforcement-learning policies; a policy that successfully deploys in MuJoCo is generally expected to transfer seamlessly to the physical environment.

2) *Learning Algorithm*: We adopt PPO with an asymmetric actor-critic architecture [23]. This variant of the standard actor-critic framework employs separate networks for the actor and the critic, permitting independent updates [24], [25]. Building upon this, we further introduce a Multi-Layer Perceptron (MLP) encoder for state estimation [26]. All training and experiments are conducted on a single NVIDIA GeForce RTX 4090 GPU with 24 GB of VRAM.

3) *Terrains*: Our environment is structured as an 8 m × 8 m terrain divided into 10 columns: one column of smooth slope, one of rough slope, six columns of stairs, and two columns of discrete obstacles, as shown in Fig. 3. To progressively increase the curriculum difficulty, the terrain is further split into 10 rows. Generally, the greater the diversity of obstacles encountered during training, the more robust the resulting policy becomes.

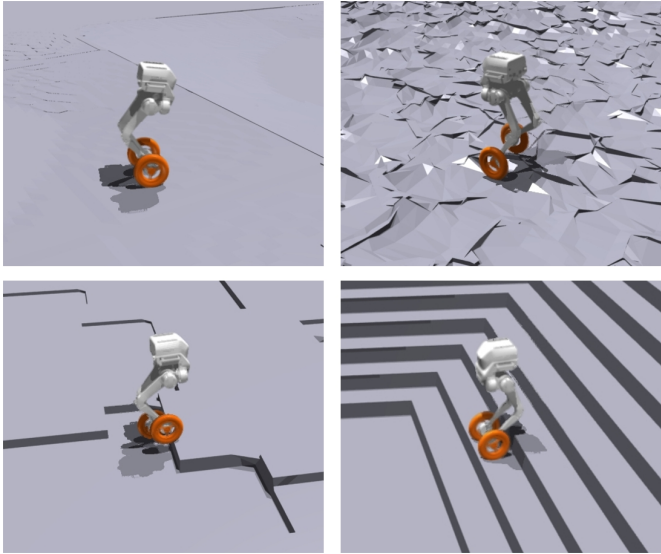


Fig. 3. Terrain type. From left to right and top to bottom: smooth slope, rough slope, discrete obstacles, and stairs.

TABLE I
OBSERVATION & PRIVILEGED INFORMATION

Symbol	Description	Units	Coeff.	Size	Noise (%)
<i>Observation (Actor / Critic)</i>					
$\vec{\theta}$	Angular velocity	rad/s	1.0	3	±20
$\vec{\gamma}$	Projected gravity	—	1.0	3	±5
$\vec{q}$	Joint positions	rad	1.0	6	±1
$\vec{\dot{q}}$	Joint velocities	rad/s	0.05	8	±50
a_{last}	Last actions	rad & rad/s	1.0	8	0
<i>Privileged Information (Critic only)</i>					
v_x	Linear velocity	m/s	2.0	3	—
$\mu_{contact}$	Avg contact forces	N	1.0	6	—
h_{height}	Height scan	m	5.0	77	—

B. Task Formulation

1) *State*: We adopt an asymmetric actor-critic architecture, so we partition the state into two parts: (i) Observations, which are accessible to both the Actor and the Critic, and (ii) Privileged Information, which is revealed to the Critic only during training. As summarized in Table I, we explicitly list the observations available to the actor during both training and deployment, as well as the privileged information reserved for the critic at training time. In particular, the term last actions denotes a composite action vector obtained by a weighted fusion of the raw actions directly output by the network and the actions derived from feedforward trajectory.

2) *Actions*: In this study, the robot’s action space has a dimension of 8. These actions correspond to the robot’s individual joints, including both leg and wheel joints. For the leg joints, the action commands are directly used as target positions for low-level proportional-derivative (PD) controllers, i.e., the controllers strive to drive the joints to these preset positions. For the wheel joints, the action vector represents target angular velocities; in other words, the joint motors operate in velocity-control mode, aiming to reach the specified

跳台和爬楼梯等任务。Chamorro 等人 [3] 证明，一种不依赖视觉或定位的无视觉强化学习策略，仅依靠位置目标、二值地形标志和非对称演员-评论家架构，即可使 Ascento 机器人攀爬15厘米高的楼梯。然而，据我们所知，尽管 Ascento 机器人的轮径为25厘米，但其仿真和硬件实验均仅展示了单级台阶的穿越能力，尚不具备攀爬窄踏板楼梯的能力。

目前，针对任意轮径的双足轮式机器人，一种通用的越障框架仍然难以实现。特别是当机器人需要在没有任何额外外部感知的情况下翻越超过其轮半径的高台阶时，现有方法往往在性能和训练收敛方面面临困难。

III. 方法论

如图2所示，我们展示了通用的接触触发式盲攀爬框架。以下章节将系统性地详细介绍训练环境、强化学习任务表述、训练流程、接触触发机制的设计、前馈指令学习的集成，以及带有具体部署细节的仿真到现实迁移策略。

A. 学习环境

1) *仿真器*：我们选择 Isaac Gym [19] 作为训练平台，因为它专为强化学习应用而设计，并配备了 GPU 加速架构，凭借其高度并行性大幅提升了智能体训练速度。此外，Isaac Gym 支持域随机化技术 [20]，通过在训练过程中引入环境变化来提高强化学习智能体的鲁棒性，从而促进策略向现实世界的迁移。为了验证我们的仿真到现实流程，我们还利用 MuJoCo [21] 对训练好的策略 [22] 进行交叉验证。MuJoCo 以其高保真仿真而闻名，被广泛用于验证强化学习策略；能够在 MuJoCo 中成功部署的策略通常预期可以无缝迁移到物理环境中。

2) *学习算法*：我们采用具有非对称演员-评论家架构的PPO [23]。这种标准演员-评论家框架的变体为演员和评论家分别使用独立的网络，允许各自独立更新 [24], [25]。在此基础上，我们进一步引入多层感知机（MLP）编码器用于状态估计 [26]。所有训练和实验均在配备24 GB显存的单个NVIDIA GeForce RTX 4090 GPU上进行。

3) *地形*：我们的环境被构建为一个8米 × 8 米的地形，分为10列：一列平滑斜坡、一列粗糙斜坡、六列楼梯和两列离散障碍物，如图3所示。为了逐步增加课程难度，地形进一步被划分为10行。一般来说，训练中遇到的障碍物多样性越大，最终得到的策略就越鲁棒。

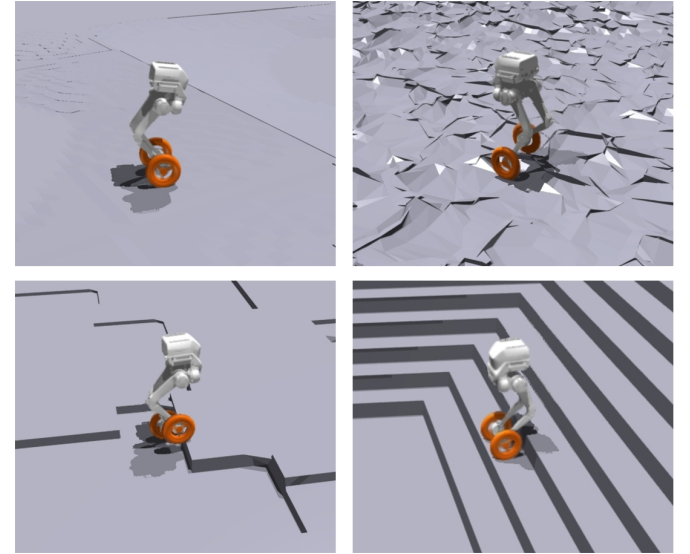


图3. 地形类型。从左到右、从上到下依次为：平滑斜坡、粗糙斜坡、离散障碍物和楼梯。

表I 观测与特权信息

符号	描述	单位	系数	Size	噪声 (%)
观测（演员 / 评论家）					
$\vec{\theta}$	角速度	弧度/秒	1.0	3	±20
$\vec{\gamma}$	投影重力	—	1.0	3	±5
$\vec{q}$	关节位置	rad	1.0	6	±1
$\vec{\dot{q}}$	关节速度	rad/s	0.05	8	±50
a_{last}	最后动作	rad 与 rad/s	1.0	8	0
特权信息（仅评论家）					
v_x	线速度	m/s	2.0	3	—
$\mu_{contact}$	平均接触力	N	1.0	6	—
h_{height}	高度扫描	m	5.0	77	—

B. 任务表述

1) *状态*：我们采用非对称演员-评论家架构，因此将状态分为两部分：(i) 观测，演员和评论家均可访问；(ii) 特权信息，仅在训练期间对评论家开放。如表I所总结，我们明确列出了演员在训练和部署期间可用的观测，以及训练时为评论家保留的特权信息。特别地，术语“最后动作”表示一个复合动作向量，该向量通过对网络直接输出的原始动作与从前馈轨迹推导出的动作进行加权融合而获得。

2) *动作*：在本研究中，机器人的动作空间维度为8。这些动作对应机器人的各个关节，包括腿部关节和轮式关节。对于腿部关节，动作指令直接用作底层比例-微分（PD）控制器的目标位置，即控制器力求将关节驱动到这些预设位置。对于轮式关节，动作向量表示目标角速度；换言之，关节电机以速度控制模式运行，旨在达到指定的

TABLE II
REWARD TERMS AND CLASSIFICATIONS

Reward	Formula	Coeff.
Task Rewards		
Lin. vel tracking x	$\exp(-20(v_{\text{cmd},x} - v_{\text{base},x})^2)$	1.2
Lin. vel tracking y	$\exp(-20(v_{\text{cmd},y} - v_{\text{base},y})^2)$	1.0
Lin. vel tracking x pb	$\frac{\Delta\phi_x}{\Delta t}$	1.0
Lin. vel tracking y pb	$\frac{\Delta\phi_y}{\Delta t}$	0.8
Ang. vel tracking	$\exp(-20 \omega_{\text{cmd}} - \omega_{\text{base}})$	1.0
Ang. vel tracking pb	$\frac{\Delta\phi\omega}{\Delta t}$	0.5
Tracking target pos	$\exp(-2\ q - q_{\text{target}}\) - 0.2\ q - q_{\text{target}}\ $	0.8
Feet air time	$\sum_i \min(t_{\text{air},i}, 0.5) \mathbb{I}_{\text{first contact},i}$	2.0
Feet contact number	$\sum_i [\mathbb{I}_{\text{contact}_i=\text{stance}_i} - 1.3 \mathbb{I}_{\text{contact}_i \neq \text{stance}_i}]$	2.0
Feet clearance	$\sum_i \mathbf{1}_{\text{swing},i} \cdot \mathbf{1}_{h_{\min} < h_i < h_{\max}}$	2.0
Style Rewards		
Nominal foot position	$\frac{1}{N} \sum_i \exp\left[-\left(\frac{(z_i - z_{\text{nom}})^2}{\sigma_z^2} + \frac{\ \mathbf{v}_{\text{cmd}}\ ^2}{\sigma_v^2}\right)\right]$	1.0
Default pose	$\sum_j q_j - q_{j,\text{default}} $	-1.0
Feet distance	$\max(0, d_{\min} - d) + \max(0, d - d_{\max})$	-10.0
Wheel zero velocity	$\exp(-\sum_{j \in \{3,7\}} \mathbf{1}_{\text{swing},j} \dot{\theta}_j^2)$	0.5
Same foot x position	$ x_0 - x_1 $	-2.0
Base height	$ h_{\text{base}} - h_{\text{target}} $	-20.0
Orientation	$\dot{g}_x^2 + \dot{g}_y^2$	-12.0
Wheel spin	$\sum_j \max(0, 0.8 r\dot{\theta}_j - \ \mathbf{v}_{\text{foot},j}\ - 0.1)$	-5.0
Opposite base vel	$\max(0, -\text{sgn}(v_{\text{cmd}})v_x)$	-40.0
Opposite wheel vel	$\sum_{j \in \{L,R\}} \max(0, -\text{sgn}(v_{\text{cmd}})\dot{\theta}_j)$	-2.0
Regularization Rewards		
Lin vel z	v_z^2	-0.3
Ang vel xy	$\omega_x^2 + \omega_y^2$	-0.01
Torques	$\sum_j \tau_j^2$	-1×10^{-5}
Dof acc	$\sum_j \ddot{q}_j^2$	-2.5×10^{-7}
Dof vel	$\sum_j \dot{q}_j^2$	-1×10^{-5}
Action rate	$\sum_j (a_j - a_j^{\text{prev}})^2$	-0.01
Action smooth	$\sum_j (a_j - 2a_j^{\text{prev}} + a_j^{\text{prev}2})^2$	-0.005
Collision	$\sum_{i \in \mathcal{S}_{\text{penalised}}} \mathbf{1}_{\ \mathbf{F}_i\ > 1}$	-50.0
Feet contact forces	$\max(0, \bar{F}_z - F_{\max})$	-5.0
Dof pos limits	$-\sum_j \max(0, q_j - q_j^{\text{limit}})$	-2.0

angular velocities.

3) *Rewards*: Our task-specific rewards are summarized in Table II. The reward function is composed of three main components:

- 1) **Task rewards**: including velocity-tracking and foot-lifting terms, which ensure the robot moves at the desired speed and follows the prescribed gait pattern.
- 2) **Style rewards**: comprising foot-pose and body-pose terms, which encourage the robot to maintain a natural and stable gait.
- 3) **Regularization rewards**: used to optimize motion smoothness and prevent superfluous joint movements.

Conditioned Reward Formulation: The rewards associated with leg-lifting maneuvers, namely Target Position Tracking, Feet Air Time, Feet Contact Number, and Feet Clearance, are formulated as conditional terms activated solely upon wheel-obstacle contact. This design effectively decouples the locomotion tasks. Specifically, it allows the robot to maintain stable and efficient high-speed wheeled cruising on flat surfaces. Meanwhile, the policy immediately triggers agile leg-lifting behaviors to negotiate obstacles once contact is detected. Con-

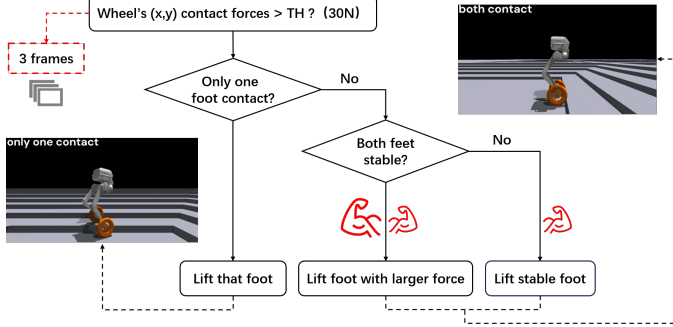


Fig. 4. Logical architecture of the contact-triggered mechanism.

sequently, the framework achieves robust obstacle traversal while fully preserving the inherent mobility of the wheeled-bipedal platform.

C. Contact-Triggered Mechanism

The methodology is primarily inspired by [17], with the core objective of updating the robot's state and determining the gait phase (stance or swing) based on the horizontal (xy -plane) contact forces measured at the feet (Fig. 4). We extend this framework with three key enhancements:

- 1) **Threshold-based Triggering**: When the contact force on either wheel exceeds a predefined threshold, the feedforward trajectory is immediately activated. This triggers an initial lift of the contacting leg, followed by a synchronized response from the contralateral leg, resulting in a coordinated alternating ascent.
- 2) **Sliding-Window Filtering**: To address the contact flickering inherent in rigid-body simulators like Isaac Gym, we adopt a three-frame sliding window to aggregate historical contact states, following the approach in [3]. To formally define the stable contact condition, we employ an indicator-based sliding window mechanism:

$$C_t = \prod_{i=0}^2 \mathbb{I}(F_{t-i} > \tau) \quad (1)$$

where $C_t \in \{0, 1\}$ denotes the binary stable contact state at time t . The term F_{t-i} represents the measured contact force magnitude in the xy -plane at the i -th preceding frame, τ is the predefined force threshold (e.g., 30N), and $\mathbb{I}(\cdot)$ is the indicator function which equals 1 if the condition is satisfied and 0 otherwise. This filtering mechanism effectively suppresses high-frequency noise while ensuring the reliability of genuine contact triggers.

- 3) **Wheel-Leg Synergetic Integration**: The triggering mechanism is tightly coupled with the rolling wheel dynamics, facilitating seamless transitions between continuous rolling and discrete stepping. This integration optimizes the trade-off between energy efficiency and high-performance obstacle traversal.

The triggering mechanism determines the lifting sequence by continuously monitoring the contact forces on both feet in real time. For each foot, the system stores the latest three

表II 奖励项与分类

奖励	公式	系数
任务奖励		
线速度跟踪x	$\exp(-20(v_{\text{cmd},x} - v_{\text{base},x})^2)$	1.2
线速度跟踪y	$\exp(-20(v_{\text{cmd},y} - v_{\text{base},y})^2)$	1.0
线速度跟踪x前馈	$\frac{\Delta\phi_x}{\Delta t}$	1.0
线速度跟踪y前馈	$\frac{\Delta\phi_y}{\Delta t}$	0.8
角速度跟踪	$\exp(-20 \omega_{\text{cmd}} - \omega_{\text{base}})$	1.0
角速度跟踪问题	$\frac{\Delta\phi\omega}{\Delta t}$	0.5
目标位置跟踪	$\exp(-2\ q - q_{\text{target}}\) - 0.2\ q - q_{\text{target}}\ $	0.8
足部腾空时间	$\sum_i \min(t_{\text{air},i}, 0.5) \mathbb{I}_{\text{first 接触},i}$	2.0
足部接触次数	$\sum_i [\mathbb{I}_{\text{contact}_i=\text{支撑相}} - 1.3 \mathbb{I}_{\text{contact}_i \neq \text{支撑相}}]$	2.0
足部离地间隙	$\sum_i \mathbf{1}_{\text{swing},i} \cdot \mathbf{1}_{h_{\min} < h_i < h_{\max}}$	2.0
风格奖励		
标称足部位置	$\frac{1}{N} \sum_i \exp\left[-\left(\frac{(z_i - z_{\text{标称}})^2}{\sigma_z^2} + \frac{\ \mathbf{v}_{\text{指令}}\ ^2}{\sigma_v^2}\right)\right]$	1.0
默认姿态	$\sum_j q_j - q_{j,\text{default}} $	-1.0
足部距离	$\max(0, d_{\min} - d) + \max(0, d - d_{\max})$	-10.0
轮子零速度	$\exp(-\sum_{j \in \{3,7\}} \mathbf{1}_{\text{swing},j} \dot{\theta}_j^2)$	0.5
相同足部x位置	$ x_0 - x_1 $	-2.0
底座高度	$ h_{\text{base}} - h_{\text{target}} $	-20.0
朝向	$\dot{g}_x^2 + \dot{g}_y^2$	-12.0
车轮旋转	$\sum_j \max(0, 0.8 r\dot{\theta}_j - \ \mathbf{v}_{\text{foot},j}\ - 0.1)$	-5.0
反向底座速度	$\max(0, -\text{sgn}(v_{\text{cmd}})v_x)$	-40.0
反向车轮速度	$\sum_{j \in \{L,R\}} \max(0, -\text{sgn}(v_{\text{cmd}})\dot{\theta}_j)$	-2.0
正则化奖励		
线速度z分量	v_z^2	-0.3
角速度xy	$\omega_x^2 + \omega_y^2$	-0.01
力矩	$\sum_j \tau_j^2$	-1×10^{-5}
自由度加速度	$\sum_j \ddot{q}_j^2$	-2.5×10^{-7}
自由度速度	$\sum_j \dot{q}_j^2$	-1×10^{-5}
动作变化率	$\sum_j (a_j - a_j^{\text{prev}})^2$	-0.01
动作平滑度	$\sum_j (a_j - 2a_j^{\text{prev}} + a_j^{\text{prev}2})^2$	-0.005
碰撞	$\sum_{i \in \mathcal{S}_{\text{penalised}}} \mathbf{1}_{\ \mathbf{F}_i\ > 1}$	-50.0
足部接触力	$\max(0, \bar{F}_z - F_{\max})$	-5.0
关节位置限制	$-\sum_j \max(0, q_j - q_j^{\text{limit}})$	-2.0

角速度。

3) 奖励：我们的任务特定奖励汇总于表II。奖励函数由三个主要部分组成：

- 1) **任务奖励**：包括速度跟踪和抬脚项，确保机器人以期速度移动并遵循规定的步态模式。
- 2) **风格奖励**：包括足部姿态和身体姿态项，鼓励机器人保持自然稳定的步态。
- 3) **正则化奖励**：用于优化运动平滑度并防止多余的关节运动。

条件奖励公式：与抬腿动作相关的奖励，即目标位置跟踪、足部腾空时间、足部接触次数和足部离地间隙，被设计为仅在轮子与障碍物接触时激活的条件项。这种设计有效地解耦了运动任务。具体而言，它使机器人能够在平坦表面上保持稳定高效的高速轮式巡航。同时，一旦检测到接触，策略会立即触发敏捷抬腿行为以越过障碍物。因

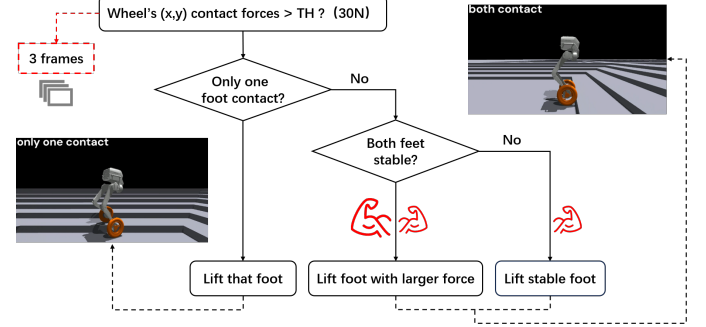


图4. 接触触发机制的逻辑架构。

此，该框架在完全保留轮式双足平台固有机动性的同时，实现了稳健的障碍物穿越。

C. 接触触发机制

该方法论主要受 [17], 启发，其核心目标是根据脚部测量的水平（ xy 平面）接触力来更新机器人状态并确定步态相位（支撑相或摆动相）（图4）。我们通过三项关键增强扩展了这一框架：

- 1) **基于阈值的触发**：当任一车轮上的接触力超过预定义阈值时，前馈轨迹立即被激活。这会触发接触腿的初始抬起，随后对侧腿产生同步响应，从而实现协调的交替上升。
- 2) **滑动窗口滤波**：为了解决Isaac Gym等刚体仿真器中固有的接触闪烁问题，我们采用三帧滑动窗口来聚合历史接触状态，遵循 [3]中的方法。为正式定义稳定接触条件，我们采用基于指示器的滑动窗口机制：

$$C_t = \prod_{i=0}^2 \mathbb{I}(F_{t-i} > \tau) \quad (1)$$

其中 $C_t \in \{0, 1\}$ 表示时间 t 时的二元稳定接触状态。术语 F_{t-i} 表示第 i 个前序帧在 xy 平面上的实测接触力大小， τ 为预定义的力阈值（例如30N）， $\mathbb{I}(\cdot)$ 为指示函数，当条件满足时取值为1，否则为0。该滤波机制有效抑制了高频噪声，同时确保了真实接触触发的可靠性。

- 3) **轮腿协同集成**：触发机制与滚动轮动力学紧密耦合，促进了连续滚动与离散步进之间的无缝切换。这种集成优化了能效与高性能障碍物穿越之间的权衡。

触发机制通过实时持续监测双脚上的接触力来确定抬升顺序。对于每只脚，系统存储最近的三帧力数据，

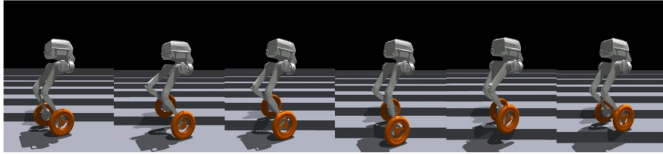


Fig. 5. When either wheel makes contact, only the contacting leg lifts.

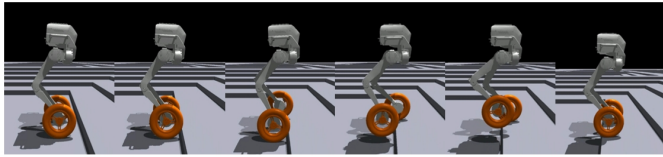


Fig. 6. If both wheels are in contact, the leg with stable contact or the larger contact force is chosen to lift.

frames of force data and designates the contact as stable contact only if all three frames exceed the threshold.

The prioritized leg-lifting decision logic is structured as follows:

- 1) **Asymmetric Contact:** If contact is detected on a single foot only, the system initiates a unilateral lift of that specific leg while the other maintains support or driving (Fig. 5).
- 2) **Bilateral Contact:** When both wheels encounter obstacles simultaneously (Fig. 6), the sequence is determined by the reliability and magnitude of the contact signals:
 - i) **Stability-First:** If only one foot satisfies the stable contact criteria (as defined by the sliding window), that foot is prioritized for lifting to ensure the maneuver is based on a genuine physical obstacle.
 - ii) **Force-Dominant:** If both feet exhibit stable contact, the system selects the leg with the greater instantaneous contact force to lift first. This minimizes the resistance torque against the base and facilitates a more agile ascent.

The mechanism determines the lifting sequence by monitoring bilateral contact forces in real-time. Specifically, the system maintains a buffer of the three most recent force frames for each foot; a "stable contact" is confirmed only if all frames in the buffer consistently exceed the threshold. This logic allows the robot to adapt its locomotion strategy dynamically to ground-truth contact conditions, fostering more natural and stable maneuvers over unstructured terrain.

D. Feedforward Instruction Learning

The concept of feedforward instruction learning, primarily inspired by [27], utilizes a baseline gait motion as a heuristic signal to guide policy exploration. In this work, we adapt this mechanism to bipedal wheeled robots by injecting feedforward trajectories exclusively into the hip-pitch and knee-pitch joints. The composite desired joint action a_t is formulated as:

$$a_t = k_{fb}a_{\pi}(t) + k_{ff}(n)a_{ff}(t), \quad (2)$$

$$a_{ff}(t) = A \left(1 - \cos \left(\frac{2\pi}{T}t \right) \right), \quad T = 0.6 \text{ s}, \quad (3)$$

where $a_{\pi}(t)$ is the neural-network policy output and $a_{ff}(t)$ is the reference trajectory.

The trajectory amplitude A and the 1:2 ratio between the hip and knee joints are chosen to initiate a basic lifting motion. While the lifting clearance is generally proportional to the trajectory amplitude, we purposefully avoid setting an excessively large value. The core objective of the feedforward signal is to serve as a behavioral guide that "seeds" the lifting primitive rather than dictating a precise geometric path. Consequently, once the robot learns the basic intent of lifting its legs, the specific clearance height is autonomously refined and optimized by the RL policy through environmental interaction.

To ensure that the final policy can autonomously execute maneuvers without external guidance during deployment, we implement a linear difference annealing schedule for the feedforward weight $k_{ff}(n)$:

$$k_{ff}(n) = \max \left(0, k_0 - n \cdot \frac{k_0}{N_{\text{ann}}} \right), \quad (4)$$

where n is the current training iteration and N_{ann} denotes the annealing horizon. As training progresses and the policy converges, k_{ff} gradually decreases to zero. At this stage, the feedforward "scaffolding" is completely removed, and the robot's motion is governed entirely by the trained network policy, ensuring a seamless transition from simulation to real-world deployment.

E. Domain Randomization

To achieve zero-shot sim-to-real transfer, we introduce a broad set of randomization factors in simulation to model real-world uncertainties and enhance the policy's generalization ability, as detailed in Table III.

In particular, we apply wide-range randomization for **friction** and **restitution** to compensate for the simplified contact physics in simulation. A broad friction range enables the blind policy to secure reliable traction across diverse real-world surfaces, while varied restitution coefficients effectively model the tires' passive damping and unpredictable energy dissipation during high-impact collisions with obstacle edges.

TABLE III
DOMAIN RANDOMIZATION

Parameter	Range	Unit
Payload mass	$[-0.5, 2]$	kg
Center of mass shift	$[-3, 3] \times [-2, 2] \times [-3, 3]$	cm
Kp Factor	$[0.8, 1.2]$	N/rad
Ka Factor	$[0.8, 1.2]$	N-s/rad
Friction	$[0.2, 1.6]$	—
Restitution	$[0.0, 1.0]$	—
Inertia	$[0.8, 1.2]$	—
Motor torque	$[0.8, 1.2]$	N
IMU offset	$[-1.2, 1.2]$	—
Default dof pos	$[-0.05, 0.05]$	rad
Step delay	$[0, 20]$	ms
Push interval	8	s
Push vel (xy)	1.0	m/s

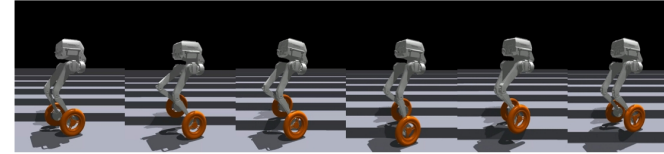


图5. 当任一车轮发生接触时，仅接触侧腿部抬升。

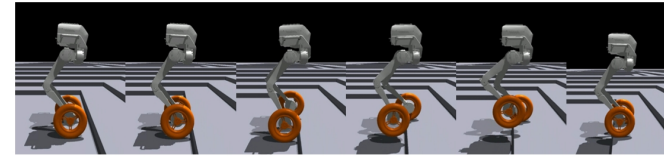


图6. 若双轮均处于接触状态，则选择具有稳定接触或接触力较大的腿部进行抬升。

并且仅当所有三帧数据均超过阈值时，才将该接触判定为稳定接触。

优先级抬腿决策逻辑的结构如下：

- 1) **非对称接触：**若仅在单脚检测到接触，系统启动该侧腿部的单侧抬升，同时另一侧保持支撑或驱动（图5）。
- 2) **双侧接触：**当双轮同时遇到障碍物时（图6），抬腿顺序由接触信号的可靠性和幅值决定：i) **稳定性优先：**若仅有一只脚满足稳定接触标准（由滑动窗口定义），则该脚优先抬升，以确保操作基于真实的物理障碍。ii) **力主导：**若双脚均表现出稳定接触，系统选择瞬时接触力较大的腿部优先抬升。这可以最小化对基座的阻力矩，并有助于更敏捷地攀越。

该机制通过实时监测双侧接触力来确定抬升顺序。具体而言，系统为每只脚维护一个包含最近三帧力数据的缓冲区；只有当缓冲区中的所有帧均持续超过阈值时，才确认“稳定接触”。这一逻辑使机器人能够根据真实接触条件动态调整其运动策略，从而在非结构化地形上实现更自然、更稳定的动作。

D. 前馈指令学习

前馈指令学习的概念，主要受 [27], 启发，利用基准步态运动作为启发式信号来引导策略探索。在本工作中，我们将这一机制适配到双足轮式机器人上，仅将前馈轨迹注入髋关节俯仰和膝关节俯仰关节。复合期望关节动作 a_t 的公式如下：

$$a_t = k_{fb}a_{\pi}(t) + k_{ff}(n)a_{ff}(t), \quad (2)$$

$$a_{ff}(t) = A \left(1 - \cos \left(\frac{2\pi}{T}t \right) \right), \quad T = 0.6 \text{ s}, \quad (3)$$

其中 $a_{\pi}(t)$ 是神经网络策略输出， $a_{ff}(t)$ 是参考轨迹。

选择轨迹幅值 A 以及髋关节与膝关节之间的1:2比例来启动基本的抬升运动。虽然抬升间隙通常与轨迹幅值成正比，但我们有意避免设置过大的数值。前馈信号的核心目标是作为行为引导，为抬升基元“播种”意图，而非规定精确的几何路径。因此，一旦机器人学会了抬腿的基本意图，具体的间隙高度将由强化学习策略通过环境交互自主细化并优化。

为确保最终策略在部署期间无需外部引导即可自主执行动作，我们对前馈权重 $k_{ff}(n)$ 实施线性差分退火调度：

$$k_{ff}(n) = \max \left(0, k_0 - n \cdot \frac{k_0}{N_{\text{ann}}} \right), \quad (4)$$

其中 n 为当前训练迭代次数， N_{ann} 表示退火范围。随着训练推进和策略收敛， k_{ff} 逐渐减小至零。在此阶段，前馈“脚手架”被完全移除，机器人的运动完全由训练后的网络策略控制，确保从仿真到现实世界部署的无缝过渡。

E. 域随机化

为实现零样本仿真到现实迁移，我们在仿真中引入广泛的随机化因素，以建模现实世界的不确定性并增强策略的泛化能力，详见表III。

具体而言，我们对<风格 id='1'>摩擦力</风格>和<风格 id='3'>恢复系数</风格>进行了大范围随机化，以补偿仿真中简化的接触物理。宽广的摩擦力范围使盲策略能够在各种真实世界表面上获得可靠的牵引力，而不同的恢复系数则有效模拟了轮胎的被动阻尼以及在与障碍物边缘发生高冲击碰撞时不可预测的能量耗散。

表III 域随机化

参数	范围	Unit
有效载荷质量	$[-0.5, 2]$	kg
质心偏移	$[-3, 3] \times [-2, 2] \times [-3, 3]$	cm
Kp系数	$[0.8, 1.2]$	牛/弧度
Ka系数	$[0.8, 1.2]$	牛-秒/弧度
摩擦力	$[0.2, 1.6]$	—
恢复系数	$[0.0, 1.0]$	—
惯性	$[0.8, 1.2]$	—
电机扭矩	$[0.8, 1.2]$	N
IMU偏移	$[-1.2, 1.2]$	—
默认自由度位置	$[-0.05, 0.05]$	rad
步态延迟	$[0, 20]$	ms
推力间隔	8	s
推力速度 (xy)	1.0	m/s

IV. EXPERIMENTS

A. Simulation Experiments

To quantify the contribution of each component in the proposed CTBC method, we conducted controlled ablation experiments across four distinct configurations. To ensure statistical reliability and eliminate the influence of stochasticity, each method was trained across three independent runs using random seeds ($s = 1, 2, 3$). All policies were evaluated after 80,000 iterations under identical simulation environments and hyperparameter setups. The configurations compared are:

- **CTBC (Proposed):** Integrates the contact-triggered leg-lifting mechanism with feedforward instructions to achieve coordinated climbing.
- **CTBC w/o Feedforward:** Retains the contact-triggered logic but removes the explicit feedforward trajectory, relying solely on indirect lifting rewards for motion generation.
- **CTBC w/o Contact-Trigger:** Maintains the feedforward trajectory but lacks the contact-dependent state transition logic, resulting in repetitive or non-adaptive lifting behaviors.
- **CTBC w/o Both:** Strips away both enhancements to serve as a baseline reinforcement learning policy for performance comparison.

For both the CTBC and CTBC w/o feedforward variants, we observed that an initial longitudinal (fore-aft) leg motion serves as a behavioral heuristic that facilitates the learning of the lifting primitive. Consequently, we adopted a two-stage training curriculum:

- 1) **Stage I (Exploration):** The policy is trained without constraints on lateral or longitudinal foot placement to encourage the discovery of successful climbing maneuvers.
- 2) **Stage II (Refinement):** Building upon the pre-trained weights from Stage I, we introduce a “same foot x -position” reward to regularize the fore-aft motion, thereby ensuring a more graceful and stable robot posture.

The comparative performance of the proposed method and its variants is illustrated in Fig. 7. As shown in Fig. 7, the complete CTBC method consistently outperforms all other variants, reaching the maximum terrain level with significantly higher sample efficiency and stability across different random seeds.

- **Impact of Feedforward Trajectory:** Without the feedforward component (CTBC w/o feedforward), the agent lacks effective heuristic guidance during exploration, resulting in a slower progression of terrain levels and a restricted performance ceiling. This underscores that the feedforward signal not only significantly accelerates the policy convergence but is also essential for maximizing the robot’s obstacle-clearing capacity.
- **Impact of Contact-Triggered Mechanism:** The variant lacking the contact-triggered mechanism (CTBC w/o contact-trigger) exhibits poor scalability. Failing to dynamically switch between rolling and stepping modes



Fig. 7. Progressive terrain-level training curves for all ablation variants. The solid lines represent the mean performance across three independent random seeds ($s = 1, 2, 3$), with the shaded areas indicating the standard deviation. Step height increases from 8 cm to 20 cm while width decreases from 50 cm to 28 cm.

TABLE IV
SUCCESS RATE (%) ON STAIRS OF INCREASING HEIGHT

Ablation Experiments	Step height (cm)						
	8	10	12	15	18	20	22
CTBC (Proposed)	100	100	100	98	96	86	70
CTBC w/o feedforward	96	96	96	92	80	58	38
CTBC w/o contact-trigger	62	60	56	46	18	2	0
CTBC w/o both	46	34	28	8	4	0	0

based on real-time contact states, the robot merely executes repetitive lifting motions. This rigid behavior leads to excessive energy consumption and severely limits its adaptability to varied terrains.

- **Baseline Performance:** The baseline policy (CTBC w/o both) fails to surmount even the most fundamental obstacles. This outcome demonstrates that neither component alone is sufficient for mastering high-dimensional stair-climbing tasks, further highlighting the necessity of the synergy between feedforward guidance and contact-triggered state transitions.

The success rates presented in Table IV further quantify these observations. To ensure the robustness of the evaluation, each success rate was statistically computed across 100 parallel environments with domain randomization fully enabled, accounting for uncertainties in friction, mass, and motor characteristics. While all methods perform reasonably well on 8 cm steps, the advantage of CTBC becomes increasingly pronounced as the difficulty escalates. Notably, our method maintains a high success rate of 86% at the 20 cm training limit and remains viable (70%) even at an extreme 22 cm height. In contrast, removing the contact-trigger causes the success rate to plummet to 2% at 20 cm. This failure occurs because the robot cannot dynamically synchronize its lifting action with actual ground contact, leading to catastrophic instability on higher obstacles.

IV. 实验

A. 仿真实验

为了量化所提出的CTBC方法中每个组件的贡献，我们在四种不同配置下进行了受控的消融实验。为确保统计可靠性并消除随机性的影响，每种方法均使用随机种子（<风格 id='1'>s</风格> = 1<风格 id='4'>, </风格>2<风格 id='6'>, </风格>3）进行了三次独立训练。所有策略均在相同的仿真环境和超参数设置下经过80,000次迭代后进行评估。比较的配置如下：

- **CTBC（提出的）：**集成接触触发式抬腿机制与前馈指令，实现协调攀爬。

- **CTBC无前馈：**保留接触触发逻辑，但移除显式前馈轨迹，仅依赖间接抬腿奖励进行运动生成。

- **CTBC无接触触发：**保留前馈轨迹，但缺乏接触相关的状态转换逻辑，导致抬腿行为重复或缺乏适应性。

- **CTBC无两者：**去除两种增强机制，作为基线强化学习策略用于性能对比。

对于CTBC和CTBC无前馈两种变体，我们观察到初始的纵向（前后）腿部运动作为一种行为启发式，有助于抬升基元的学习。因此，我们采用了两阶段训练课程：

- 1) **阶段一（探索）：**策略在不对足部横向或纵向位置施加约束的情况下进行训练，以鼓励发现成功的攀爬动作。
- 2) **阶段二（精炼）：**在阶段一的预训练权重基础上，我们引入“同脚 x 位置”奖励来规范前后运动，从而确保机器人姿态更加优雅稳定。

所提出的方法及其变体的对比性能如图7所示。如图7所示，完整的CTBC方法始终优于所有其他变体，以显著更高的样本效率和跨不同随机种子的稳定性达到最大地形等级。

- **前馈轨迹的影响：**没有前馈组件（CTBC无前馈）时，智能体在探索过程中缺乏有效的启发式引导，导致地形等级进展较慢且性能上限受限。这凸显了前馈信号不仅能显著加速策略收敛，而且对于最大化机器人的越障能力至关重要。

- **接触触发机制的影响：**缺少接触触发机制的变体（CTBC无接触触发）表现出较差的可扩展性。由于无法根据实时接触状态动态切换滚动与迈步模式

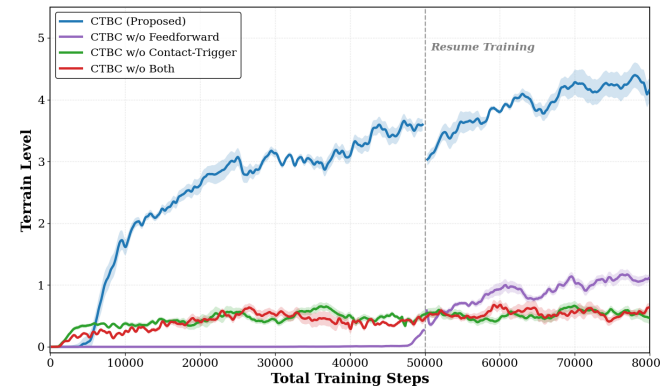


图7. 所有消融变体的渐进式地形级训练曲线。实线代表三个独立随机种子 ($s = 1, 2, 3$) 的平均性能，阴影区域表示标准差。台阶高度从8厘米增加到20厘米，而宽度从50厘米减小到28厘米。

表IV 递增高度楼梯上的成功率 (%)

消融实验	台阶高度（厘米）						
	8	10	12	15	18	20	22
CTBC（提出的）	100	100	100	98	96	86	70
CTBC无前馈	96	96	96	92	80	58	38
CTBC无接触触发	62	60	56	46	18	2	0
CTBC无两者	46	34	28	8	4	0	0

，机器人仅执行重复抬升动作。这种刚性行为导致过度能耗，并严重限制了其对多变地形的适应性。

- **基线性能：**基线策略（CTBC无两者）甚至无法跨越最基本的障碍。这一结果表明，单独任何一个组件都不足以掌握高维爬楼梯任务，进一步凸显了前馈引导与接触触发状态转换之间协同作用的必要性。

表IV中呈现的成功率进一步量化了这些观测结果。为确保评估的鲁棒性，每个成功率均在完全启用域随机化的100个并行环境中进行统计计算，充分考虑了摩擦力、质量和电机特性方面的不确定性。虽然所有方法在8厘米台阶上均表现良好，但随着难度升级，CTBC的优势愈发显著。值得注意的是，我们的方法在20厘米训练极限处仍保持86%的高成功率，即使在22厘米的极端高度下依然可行（70%）。相比之下，移除接触触发机制后，在20厘米处的成功率骤降至2%。这一失败源于机器人无法将抬腿动作与实际地面接触动态同步，导致在更高障碍上出现灾难性不稳定。

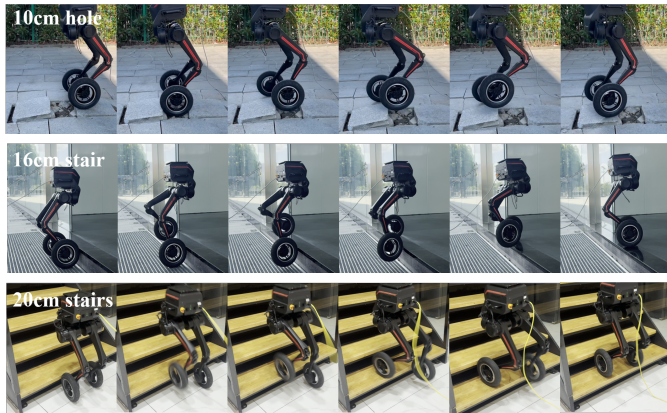


Fig. 8. Snapshots of real-world experiments on the *LimX Dynamics Tron1* robot. Top: Escaping a 10 cm deep hole through rapid contact-triggered leg lifting. Middle and Bottom: Successfully ascending 16 cm and 20 cm high stairs, respectively, demonstrating robust blind locomotion over varied obstacle heights.

B. Real-World Experiments

The CTBC policy, operating at a control frequency of 50 Hz, was deployed on the 8-DoF wheeled-legged robot, *LimX Dynamics Tron1*, without any exteroceptive sensing (e.g., LiDAR or cameras). To evaluate the robustness and sim-to-real transferability of the policy, we conducted experiments in two challenging scenarios: hole escape and stair climbing (Fig. 8).

- **Hole Escape:** Upon a wheel dropping into a 10 cm deep void, the resultant contact force immediately exceeds the trigger threshold. Despite the discretized 50 Hz control loop, the policy responds by executing a rapid lifting motion, enabling the robot to extricate itself smoothly without losing balance.
- **Stair Climbing:** When encountering 16 cm and 20 cm steps, the policy dynamically determines the lead leg based on the transient contact forces at the impact moment. This facilitates a rapid ascent while maintaining postural stability under blind conditions.

As illustrated in Fig. 9, the robot successfully ascends a series of 20 cm continuous open-gap stairs. This scenario represents a heightened level of difficulty due to the extreme discontinuity and the hollow structure of the support surfaces. Even in the absence of a solid riser, our method effectively covers these complex terrains: the contact-triggered mechanism provides the necessary precision to initiate lifting only upon impact with the narrow tread, preventing the wheels from falling into the gaps. This robust performance demonstrates that the learned policy can reliably handle repetitive, high-impact locomotion on sparse terrain.

Furthermore, to verify the cross-platform versatility of our framework, we transferred the CTBC policy to the *Cowarobot R0*, a heavy-duty bipedal wheeled-legged robot integrated with a robotic arm and 8-DoF legs. The *Cowarobot R0* presents a stark contrast to *LimX Dynamics Tron1* (which weighs less than 20 kg) with a total mass of 65 kg and significantly different hardware characteristics. Specifically, it utilizes small-diameter (11 cm) solid rubber tires instead

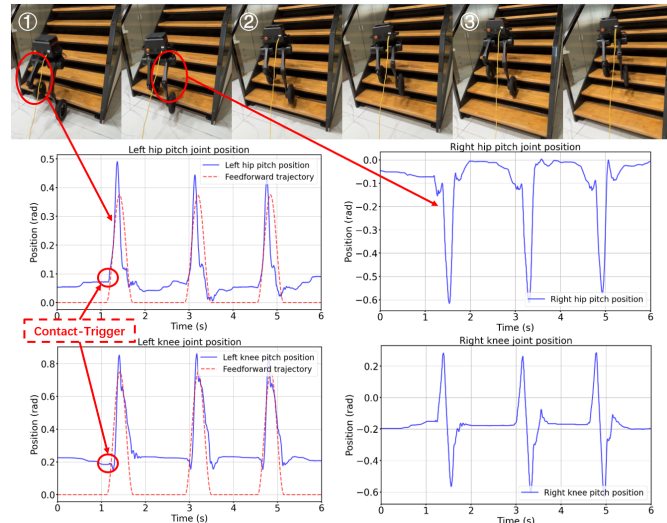


Fig. 9. Continuous blind ascent on 20 cm high open-gap stairs. The results demonstrate the policy’s precision in handling extreme surface sparsity. The real-time joint data plots confirm the effectiveness of the guidance: the executed trajectories (solid blue) effectively track the prescribed feedforward instructions (dashed red) upon being activated. Notably, the “Contact-Trigger” points mark the precise moments where the agent transitions from rolling to stepping based on impact, enabling stable clearance of consecutive hollow treads.

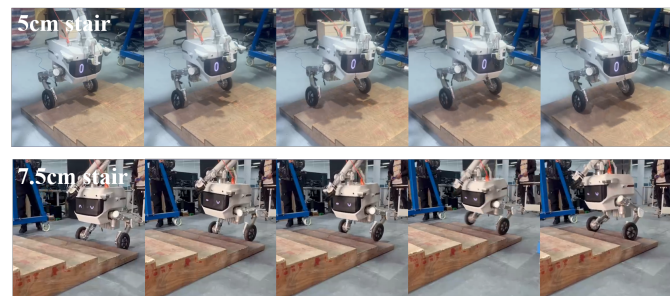


Fig. 10. *Cowarobot R0* ascending 5 cm (top) and 7.5 cm (bottom) steps. Despite the 65 kg mass and low joint velocity (< 2 rad/s), the robot achieves continuous ascent without fine-tuning, demonstrating the zero-shot transferability of CTBC across platforms with distinct scale and dynamics.

of the pneumatic ones used on *LimX Dynamics Tron1*, and its maximum knee joint velocity is constrained to within 2 rad/s. Despite these substantial discrepancies in physical scale, contact dynamics, and actuator bandwidth, the robot successfully achieved continuous ascent of 5 cm and 7.5 cm steps without any fine-tuning (as shown in Fig. 10). Such successful deployment on a high-inertia platform, characterized by distinct contact dynamics and stringent kinematic constraints, strongly underscores the exceptional robustness and generalization capabilities of the proposed method.

Remarkably, the policy exhibits exceptional generalization: even if the annealing schedule is bypassed and the feedforward signal is abruptly removed, the robot maintains its ability to surmount 20 cm steps. This confirms that the neural network has successfully internalized the lifting maneuver, transitioning from guided exploration to autonomous execution. Furthermore, although the feedforward trajectory was originally designed for a 10 cm lift height, the learned policy successfully

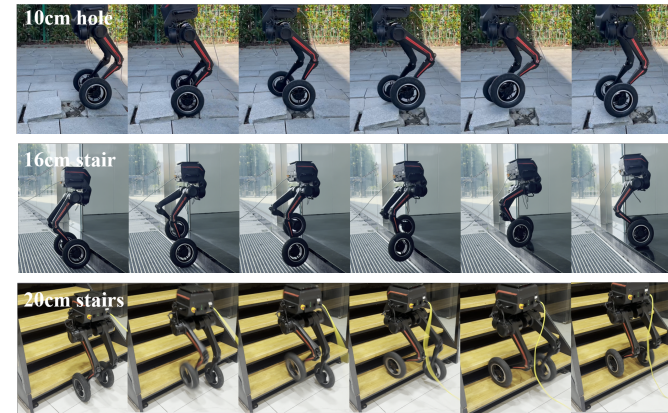


图 8. 在 *LimX Dynamics Tron1* 机器人上进行的真实世界实验快照。上图：通过快速接触触发式抬腿动作，从 10 厘米深的坑洞中脱困。中图和下图：分别成功攀爬 16 厘米和 20 厘米高的楼梯，展示了在不同障碍物高度下稳健的盲态运动能力。

B. 真实世界实验

CTBC策略以50 Hz的控制频率部署在8自由度轮腿机器人*LimX Dynamics Tron1*上，无需任何外部感知（如激光雷达或摄像头）。为评估策略的鲁棒性及仿真到实物的迁移能力，我们在两个具有挑战性的场景中进行了实验：脱坑和爬楼梯（图8）。

- **脱坑：**当轮子落入 10 厘米深的空洞时，产生的接触力会立即超过触发阈值。尽管控制回路为离散的 50 Hz，策略仍能通过执行快速抬升运动做出响应，使机器人能够平稳脱困而不失去平衡。
- **爬楼梯：**当遇到 16 厘米和 20 厘米高的台阶时，策略会根据撞击瞬间的瞬态接触力动态确定引导腿。这有助于在盲态条件下快速攀爬，同时保持姿态稳定性。

如图 9 所示，机器人成功攀爬了一系列 20 厘米高的连续开放式间隙楼梯。由于支撑面的极度不连续性和中空结构，这一场景代表了更高的难度。即使在缺乏实心踢脚板的情况下，我们的方法也能有效覆盖这些复杂地形：接触触发机制提供了必要的精度，仅在撞击窄踏板时启动抬升动作，从而防止轮子落入间隙中。这一稳健的表现证明，学习到的策略能够可靠地应对稀疏地形上重复的高冲击运动。

此外，为验证我们框架的跨平台通用性，我们将 CTBC策略迁移至*Cowarobot R0*——一款集成机械臂和8自由度腿部、总质量达65 kg的重型双足轮腿机器人。*Cowarobot R0*与*LimX Dynamics Tron1*（重量不足20 kg）形成了鲜明对比，其硬件特性也显著不同。具体而言，它采用直径较小（11 cm）的实心橡胶轮胎，而非

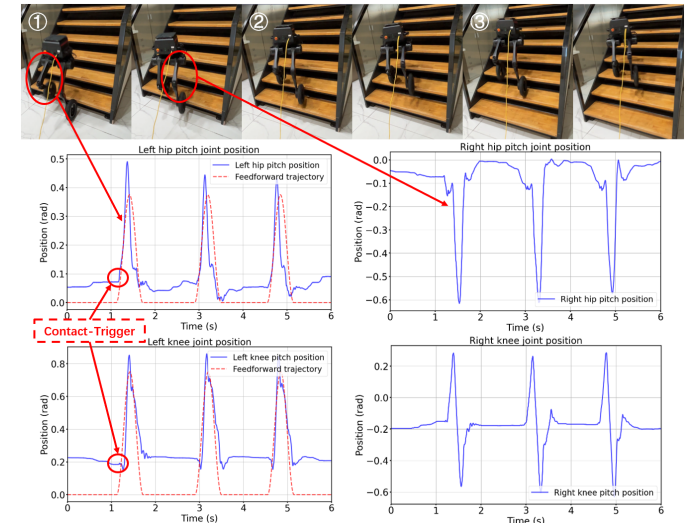


图9. 在20厘米高的开放式间隙楼梯上连续盲攀。结果表明，该策略在处理极端表面稀疏性方面具有精确性。实时关节数据图证实了引导的有效性：执行轨迹（蓝色实线）在激活后有效跟踪预设的前馈指令（红色虚线）。值得注意的是，“接触触发”点标志着智能体基于冲击从滚动过渡到步行的精确时刻，从而能够稳定跨越连续的镂空踏板。

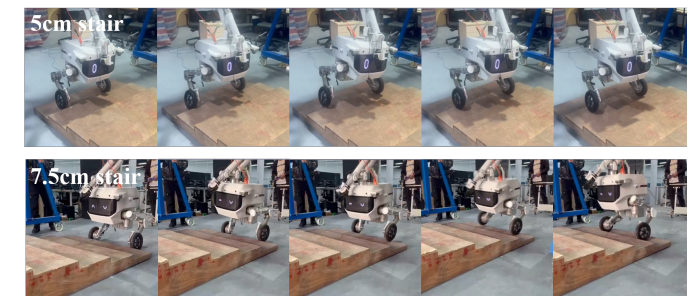


图10. *Cowarobot R0* 攀爬5厘米（顶部）和7.5厘米（底部）台阶。尽管具有65千克的质量和较低的关节速度 (< 2 rad/s)，该机器人无需微调即可实现连续攀爬，展示了CTBC在具有不同尺度和动力学的平台上的零样本迁移能力。

*LimX Dynamics Tron1*上使用的充气式轮胎，且其最大膝关节速度被限制在2 rad/s以内。尽管在物理尺度、接触动力学和执行器带宽方面存在显著差异，该机器人仍成功实现了对5 cm和7.5 cm台阶的连续攀爬，且无需任何微调（如图10所示）。这种在高惯性平台上的成功部署，加之其独特的接触动力学和严苛的运动学约束，有力地彰显了所提出方法卓越的鲁棒性和泛化能力。

值得注意的是，该策略展现出卓越的泛化能力：即使绕过退火调度并突然移除前馈信号，机器人仍能保持跨越20厘米台阶的能力。这证实了神经网络已成功内化抬升动作，从引导式探索过渡到自主执行。此外，尽管前馈轨迹最初是为10厘米抬升高度设计的，学习到的策略仍能成功

scales this behavior adaptively to clear 20 cm obstacles. For even more challenging terrains, the framework remains highly extensible; by increasing the feedforward amplitude and the reward-constrained lifting range, the robot can acquire even more powerful climbing capabilities through re-training.

V. CONCLUSIONS

In conclusion, this paper presents a contact-triggered, blind locomotion framework for bipedal wheeled-legged robots, successfully bridging the gap between prescribed feedforward guidance and adaptive reinforcement learning. By internalizing motion primitives through a two-stage training scheme, the robot achieves robust stair-climbing and hole-traversal without any exteroceptive sensing. Experimental validations on both the 20 kg *LimX Dynamics Tron1* and the 65 kg *Cowarobot R0* demonstrate the framework’s exceptional cross-hardware versatility. Despite significant discrepancies in physical scale, wheel diameters, and tire materials (pneumatic vs. solid rubber), the policy generalizes effectively to obstacles multiple times the height of the initial guidance.

Nevertheless, certain limitations remain: the extended gait in the first training phase restricts the contact frequency of the rear legs, leading to a persistent kinematic bias where the policy favors leading with the front legs. Furthermore, the current system remains purely blind, limiting its efficiency in complex, unstructured environments. Future work will explore symmetry-breaking rewards to eliminate gait bias and integrate this robust blind policy as a low-level reactive controller within a hierarchical perceptive architecture. This synergy between ”blind-reflex” and ”vision-based planning” will ultimately empower bipedal wheeled-legged robots with fully autonomous navigation and traversal capabilities in unknown and hazardous terrains.

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自适应地扩展这一行为以跨越20厘米的障碍物。对于更具挑战性的地形，该框架仍具有高度可扩展性；通过增加前馈幅度和奖励约束的抬升范围，机器人可以通过重新训练获得更强大的攀爬能力。

五、结论

总之，本文提出了一种面向双足轮腿机器人的接触触发式盲态运动框架，成功弥合了预设前馈引导与自适应强化学习之间的鸿沟。通过两阶段训练方案将运动基元内化，该机器人在无需任何外部感知的情况下实现了稳健的爬楼梯和越障能力。在20公斤的*LimXDynamics Tron1* 和65公斤的*CowarobotR0* 上进行的实验验证展示了该框架卓越的跨硬件通用性。尽管在物理尺寸、轮径和轮胎材料（充气式与实心橡胶）方面存在显著差异，该策略仍能有效泛化到数倍于初始引导高度的障碍物上。

然而，仍存在一定的局限性：第一阶段训练中的延长步态限制了后腿的接触频率，导致策略倾向于前腿先行的持续性运动学偏差。此外，当前系统仍为纯盲态，限制了其在复杂非结构化环境中的效率。未来工作将探索对称性破缺奖励以消除步态偏差，并将这一稳健的盲策略作为低级反应控制器集成到分层感知架构中。这种“盲态反射”与“基于视觉的规划”之间的协同作用，最终将使双足轮腿机器人在未知和危险地形中具备完全自主导航与穿越能力。

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